

Product Carbon Footprint Study Report

Product Name: light emitting diode

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: light emitting diode

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg light emitting diode

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg light emitting diode.

2. System Boundary

light emitting diode production in China

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of light emitting diode produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **236.224 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

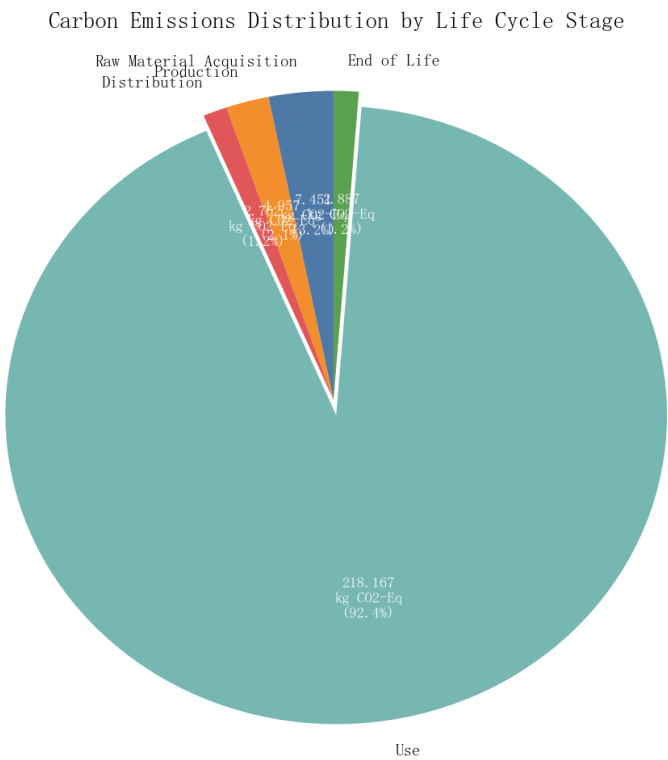


Table 1: Carbon Emissions by Life Cycle Stage

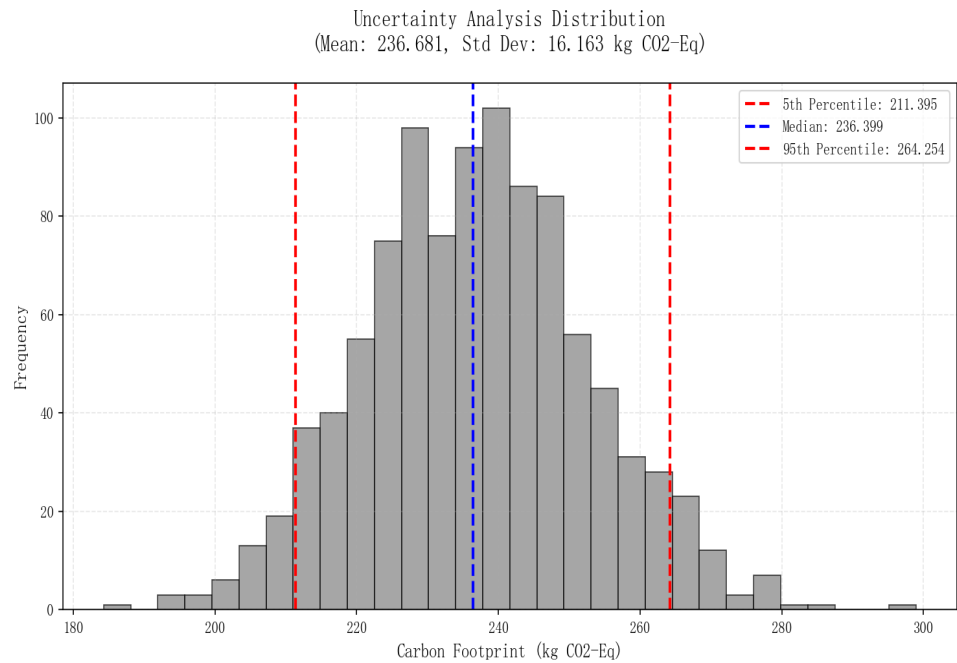
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	7.451	3.15
Production	4.957	2.10
Distribution	2.762	1.17

Use	218.167	92.36
End of Life	2.887	1.22
Total	236.224	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	236.681
Standard Deviation	16.163
Coefficient of Variation (CV)	6.83%
Median	236.399
5th Percentile	211.395
95th Percentile	264.254

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the analyzed light-emitting diode (LED) product is 236.224 kg CO₂-Eq. This value reflects the cumulative greenhouse gas emissions across all lifecycle stages, which is considered moderate for an electrical product given its functional unit and service life. The analysis reveals a highly skewed contribution pattern, with the use phase dominating at 92.4% (218.167 kg CO₂-Eq). This indicates that the product's environmental impact is overwhelmingly driven by electricity consumption during operation, rather than material extraction, manufacturing, or disposal.

The raw material extraction (3.2%), production (2.1%), distribution (1.2%), and end-of-life (1.2%) stages contribute minimally in comparison. Notably, the distribution and end-of-life phases each account for only 1.2% of the total, suggesting that logistics and disposal have negligible impacts relative to use. The major emission source is therefore the energy mix used during the product's operational lifetime. Given that LEDs are known for high energy efficiency, this result highlights that overall impact reduction must focus on extending product lifespan, improving energy efficiency, or greening the electricity grid. The top process contributions are listed as unknown, indicating a need for better data granularity in the model.

2. Statement of Assumptions and Limitations

Key assumptions in this study include a fixed operational lifespan and a constant electricity consumption rate for the LED during the use phase. The carbon footprint of electricity was likely based on a regional or national average grid mix, which may not reflect local variations or future decarbonization trends. Additionally, it was assumed that end-of-life treatment follows standard recycling or landfill practices without crediting for material recovery. These assumptions simplify modeling but can influence the accuracy of results.

Data limitations are evident from the "Unknown" entries in top process contributions, suggesting that specific production processes or materials were not adequately characterized in the background database. The coefficient of variation (CV) of 6.83% indicates moderate uncertainty, with a 90% confidence interval from 211.395 to 264.254 kg CO₂-Eq. This uncertainty likely stems from variability in electricity sources, manufacturing yields, and transportation distances. The system boundary is "cradle-to-grave," covering raw material, production, distribution, use, and end-of-life stages, but excludes capital equipment, maintenance, and infrastructure impacts. Such restrictions may underestimate total environmental burdens, particularly for supply chain emissions not captured.

3. Suggestions for Improvement

- ****Improve energy efficiency****: Prioritize design optimizations to reduce the LED's power consumption during the use phase. Since this stage accounts for over 92% of emissions, even a 10% reduction in wattage can yield significant carbon savings over the product's lifetime. Pair with smart controls or dimming features to further minimize operational energy use.

- ****Promote renewable electricity****: Encourage the adoption of green energy tariffs or on-site solar for the product's typical operating environment. For large-scale users (e.g., commercial lighting), this can directly cut the use-phase carbon footprint by up to 80%, depending on grid mix.
- ****Extend product lifespan****: Improve durability and modular design to enable easy replacement of LED components (e.g., drivers) rather than whole units. A longer operational life reduces the per-year carbon footprint and amortizes manufacturing impacts more effectively.
- ****Enhance data transparency****: Partner with suppliers to obtain primary data for key raw materials and production steps, replacing "Unknown" process entries. This will improve the reliability of the LCA model and allow targeted hot-spot analysis beyond the use phase.
- ****Implement circular economy practices****: Design for easier disassembly and material recovery at end of life. While the end-of-life stage currently has low impact, recovering rare metals and reducing landfilling can yield additional environmental and economic benefits.